

Generic aspects of multimedia presentation

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The paper will identify generic aspects of multimedia presentation which are common to a wide range of services and applications (e.g. user interaction, spatial and temporal synchronisation, data referencing and control structures). These characteristics will be presented in a network, platform and service independent manner and with particular reference to the emerging ISO MHEG multimedia coding standard. In addition, the paper will consider the ways in which such a standard may affect the architecture of future multimedia platforms.

1. Introduction

The rapid development of multimedia computing has, to date, been driven by the impact of new hardware and software technology in fields such as storage, compression, networks and computing power. Although this trend is continuing apace, it is only recently that technology has matured to the extent that mass-market applications are becoming viable.

Traditionally, multimedia applications have been targeted at niche markets such as training, corporate presentation and desk-top publishing. Technology solutions in these areas have been diverse, non-standardised and have failed to provide integrated and stable platforms for application development. However, ongoing technological change and standardisation have moved the focus of multimedia applications towards the mass market and in doing so have changed the emphasis of development. Tomorrow's multimedia platforms will be consumer items aimed at the residential 'lounge' environment and will offer a large user base together with a stable and well-integrated platform for the deployment of new and innovative multimedia services. In particular, the recent emergence of the multimedia PC equipped with CD-ROM, sound card and video capability has highlighted the potential consumer interest in multimedia products. Ongoing development of such technology is expected to include the convergence of PC, TV and communications services within general-purpose devices such as TV set-top boxes (although the extent and nature of this convergence is still hotly debated) [1].

With the emergence of mass-market multimedia systems, it is increasingly apparent that the value of a multimedia application will be perceived not as a function

of the technology platform, but more as a function of the information available through the application. In this regard it is clear that content provision (in particular, aspects of production, and copyright), traditionally the domain of the entertainment industry, will be a major source of both costs and revenue. Hence, the production of content that is reusable across applications and services will be a major factor in the reduction of costs and in increasing profits.

Content production and reuse impacts on a number of key technology areas, such as multimedia hardware platforms, authoring systems, storage and distribution techniques and media-encoding formats. This paper focuses on one particular area, the representation of complex multimedia content which may be reused across applications and services. The paper examines the reasons why a standardised approach for representing multimedia content is desirable, and discusses the concepts supported by such an approach with reference to the emerging ISO MHEG (Multimedia and Hypermedia Information Coding Expert Group) standard [2]. In addition, the paper briefly illustrates the nature of future multimedia architectures based around reusable multimedia content.

2. The need for standardised multimedia coding

A simplified multimedia architecture is shown in Fig 1. This architecture is typical of historical multimedia systems in which a vertical segmentation of functionality is present throughout the system. In such a system inter-application communication is either impossible or requires much effort to transcode between data formats, media encodings and command sets which are uniquely defined within each

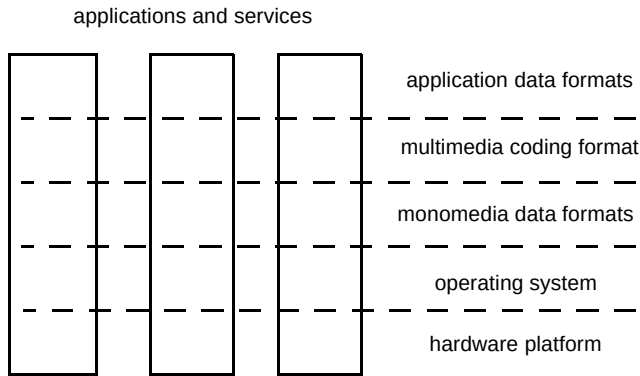


Fig 1 Segmentation in typical multimedia systems.

application context. Furthermore, in such a scenario, hardware environments are often widely incompatible making delivery of an application across multiple platforms costly.

The advent of standardised hardware platforms and operating systems such as the multimedia PC has improved interworking of applications by providing a stable and relatively mature hardware base around which applications may be developed. This has been the first stage in the rationalisation of multimedia systems and a marked step towards inter-application operation.

The next step on the road to application interworking has been the development of standardised monomedia encodings. Typically such developments have been the role of *de jure* standards bodies such as ISO (International Standards Organisation) and the ITU (International Telecommunications Union), and have resulted in technologies such as MPEG video compression (Moving Picture Experts Group), JPEG image compression (Joint Photographic Experts Group) and other media encodings. Furthermore, many proprietary encodings have also been developed or are used for historic reasons. Typically, proprietary systems are aimed at specific hardware and software platforms (e.g. PC running Microsoft Windows™).

However, it is only relatively recently that the need for standardised multimedia coding schemes has been recognised. The advent of hardware capable of supporting and presenting complex multimedia content consisting of multiple media types (text, image, audio, video) has highlighted the requirement for reuse not only of monomedia data but also the multimedia structures that surround such data. Indeed, it is clear that considerable effort, and hence value, is invested in the production of a multimedia presentation.

A successful multimedia coding scheme should support facilities applicable to a wide variety of applications and services and should result in widespread reuse of

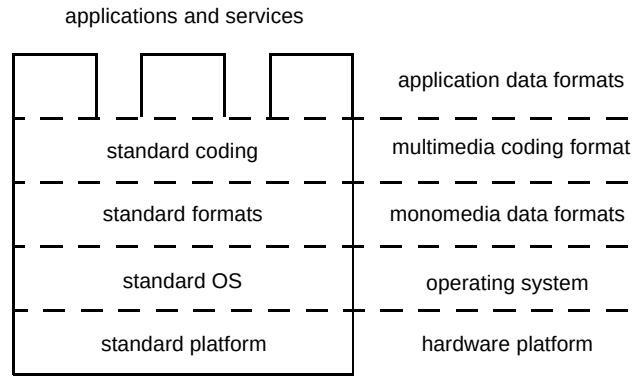


Fig 2 Standard monomedia and multimedia data formats.

multimedia intellectual property. In this scenario, the simplified architecture is as illustrated in Fig 2.

3. Techniques for multimedia coding

At present a wide variety of techniques are available for creating and representing multimedia content ranging from traditional DTP packages, 4GL programming languages and fast prototyping languages through to complex middleware with associated multimedia scripting tools, databases, presentation and storage sub-systems. However, many of these systems blur the distinction between multimedia representation and the application, resulting in the production of multimedia content which is either fully encapsulated within a specific application or is not reusable across applications. Furthermore, multimedia content produced by these tools is typically coded in a proprietary format and hence is inextricably bound to particular software sub-systems and hardware platforms.

The problem of producing a non-proprietary multimedia content encoding, which is supportable across a wide variety of platforms, has recently been addressed within the ISO MHEG standards body. The aim of this body has been to produce a coding scheme for multimedia information which supports:

- application generic multimedia presentation concepts,
- fully self-defined complex multimedia presentations,
- multimedia content reuse,
- independence of application context,
- independence of hardware and software platform,
- independence of distribution mechanism,
- presentation on restricted resource hardware platforms,
- established monomedia coding formats,
- an open coding scheme.

The initial phase of this work is now nearing completion and MHEG Part 1 (Coded Representation of Multimedia and Hypermedia Information Objects) is expected to be published in late 1995. Extensions to this work will consider the migration of MHEG concepts into a fully functional multimedia scripting language supporting higher level application logic.

4. The MHEG model

The MHEG model is a simple one — there should exist a standardised coding scheme for multimedia data structures which can be incorporated into diverse applications, services and platforms. The coding scheme should offer facilities to represent concepts which have been identified as being generic to a wide variety of applications and that are sufficiently flexible to support complex multimedia presentation. In addition, the coding scheme should be applicable to a wide range of platforms, storage and distribution techniques with a particular emphasis on presentation by restricted resource devices (e.g. TV set-top boxes) and distribution via telecommunications networks. The MHEG Part 1 Standard provides a public specification for the multimedia data structures developed using ASN.1 (Abstract Syntax Notation 1). The ASN.1 notation provides a specification of complex nested data structures which may be encoded to a bit stream according to standard rules. These bit streams may be interpreted by any general-purpose computer equipped with an ASN.1 decoder and hence act as an intermediate transfer format for multimedia objects between different internal coding schemes. The basic MHEG model is illustrated in Fig 3.

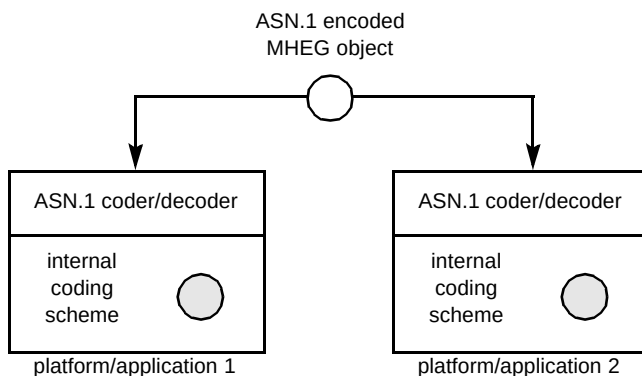


Fig 3 Basic MHEG model.

Figure 3 shows that multimedia data structures held within platforms and applications may be encoded in a private or proprietary format applicable to the particular processing environment. Reuse and platform/application interworking is achieved by encoding internal multimedia formats to and from the standard MHEG format. It is essential therefore that the MHEG coding scheme is sufficiently flexible to support most application requirements for multimedia presentation.

It should be noted that the MHEG model does not assume any particular set of presentation, processing, storage or distribution techniques for the encoded multimedia data.

5. Generic concepts in multimedia coding

The following sections detail some of the key concepts of multimedia presentation which should be supported by a generic multimedia coding scheme.

As mentioned previously, existing multimedia coding schemes are in general inextricably bound within application or platform contexts. They may offer a subset of these key concepts or entirely different approaches to multimedia presentation. However, it is believed that the concepts presented here are sufficient to support a wide range of typical multimedia applications.

5.1 Multimedia object life cycles

During its lifetime, a multimedia object may exist in a variety of states. These states and possible changes between them are illustrated in Fig 4.

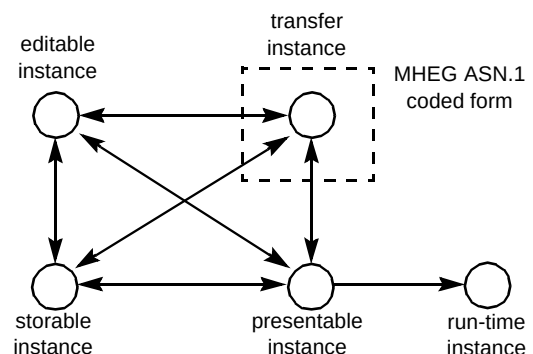


Fig 4 Multimedia object life cycle.

The states of a multimedia object are fivefold.

- The editable instance, existing in a possibly private format within an authoring or editing application — the coding scheme employed should be amenable to fast revision, update and presentation and will be compatible with local programming tools. The MHEG coded form is not generally considered suitable for editing.
- The storable instance, existing in a possibly private format within a storage medium — the MHEG coded form is considered suitable for storage purposes and is independent of storage media.

- The transfer instance, existing between applications and platforms — the MHEG standard's primary function is to provide a standard encoding for transfer.
- The presentable instance, existing in a possibly private format within a presentation device (e.g. PC, set-top box) — the coding scheme employed should be amenable to fast presentation and will be compatible with the local programming environment and presentation devices. The MHEG coded form is not generally considered suitable for presentation.
- The run-time instance, existing in a possibly private format within a presentation device — the run-time instance represents an individual presentation to the user of the presentable instance. Many run-time instances may exist (simultaneously) as representations of a single presentable instance, sharing the monomedia and multimedia structures contained in the presentable instance. The coding format employed should be amenable to fast presentation and compatible with the local programming environment and presentation devices. The MHEG coded form is not generally considered suitable for presentation.

A multimedia coding scheme should address the migration of a multimedia object between the various states and in particular between the storage, transfer, presentable and run-time states within an application. Sophisticated referencing mechanisms are required to support the interaction between these states (see section 5.4).

The MHEG standard deals explicitly with the transfer, presentable and run-time states and adopts the model illustrated in Fig 5.

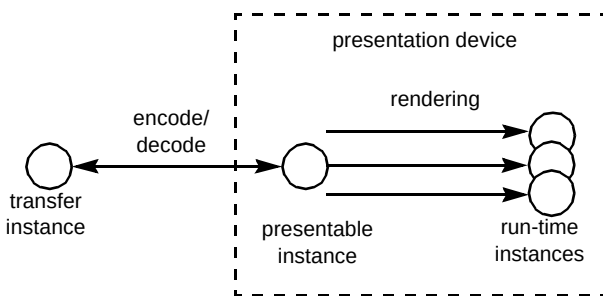


Fig 5 MHEG object life cycle.

In this model, the transfer instance (encoded in MHEG form) is delivered to the presentation device. It is then the responsibility of the presentation device to decode the transfer instance to a suitable internal form, the presentable instance, ready for presentation. Many presentations of the presentable instance, the run-time instances, may then be made by rendering the monomedia data and multimedia structures expressed within the presentable instance.

The MHEG standard dictates how objects encoded according to the MHEG standard should be rendered by the presentation device.

5.2 Composition

A key function of a multimedia coding scheme is to describe the composition of individual media objects (text, image, graphics, video, etc) into complex multimedia structures and to control the relationships between the elements of these structures in terms of their presentation to and interaction with a user. This is illustrated in Fig 6.

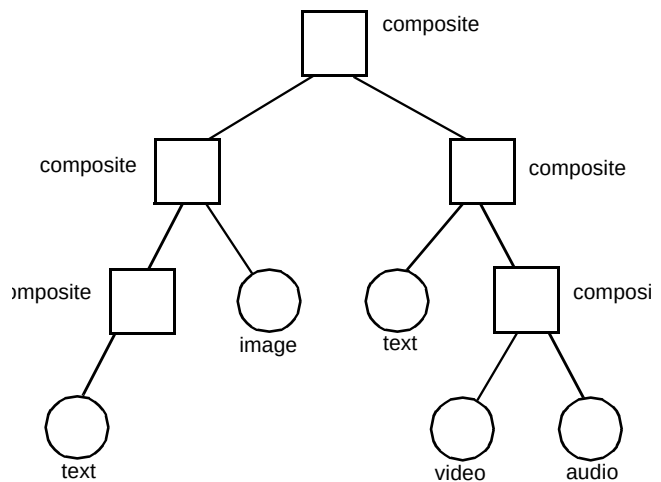


Fig 6 Composition of multimedia structures.

Such structures may become arbitrarily complex with multiple levels of composition. Each element within the structure represents a reusable entity which (with or without its sub-components) may be transplanted to other structures. The basic requirement for multimedia content reuse is thus satisfied within the scope of the coding scheme.

The MHEG standard supports composition through the composite object structure, which is used to group sets of component objects and to express the rules and relationships governing their presentation and interaction.

In order to support composition and the meaningful presentation of multimedia content to a user, a generic multimedia coding scheme must offer a range of referencing, control, presentation and interaction facilities. These facilities may be used in the authoring stage to define multimedia content in terms of simpler monomedia and multimedia objects.

In particular, the following four facilities are required.

Support for multiple media types

The coding scheme should support the referencing and/or encapsulation of objects encoded according to existing monomedia encoding formats. These objects comprise the basic visible and audible elements of a multimedia presentation. Typically encoding formats are available for monomedia data such as video (MPEG, H.261, etc), image (JPEG, GIF, TIFF, BMP, etc) and audio (G.711, WAV, etc). However, consideration should also be given to support for complex media types such as data sets (spreadsheets, CAD files) and structured text (DTP, word processor output).

Two ways of integrating monomedia objects within a coded structure are illustrated in Fig 7 — encapsulation and referencing. Encapsulation allows encoded media data to be carried within the monomedia object representing the data enabling the distribution of the monomedia object as a single entity. Referencing allows encoded media data to be stored or distributed separately from the monomedia object. This is of particular use where the referenced media data is large and may not be required for presentation or is to be streamed in real time from a data source (e.g. video camera).

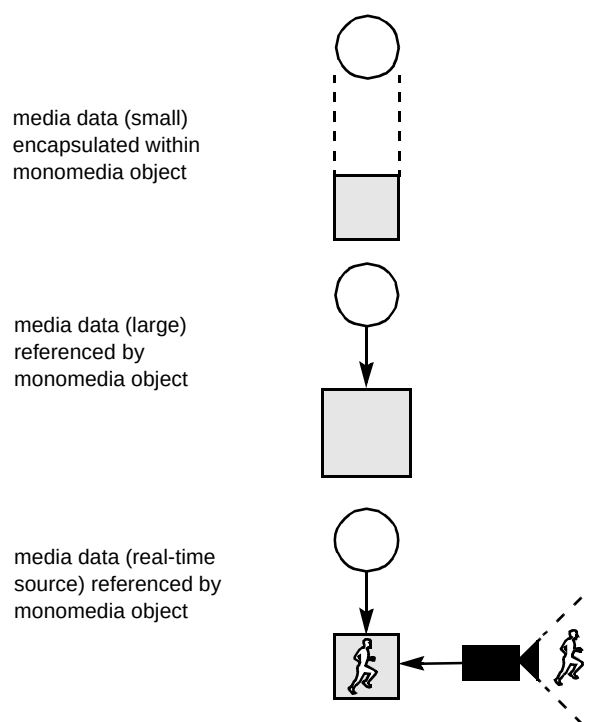


Fig 7 Encapsulation and referencing of media data.

Additionally, the multimedia coding scheme should be able to uniquely identify each encoding to the presentation sub-system in order to ensure that the correct resources and devices are available for presentation of the media data. A standardised identifier for each data encoding is thus

desirable to ensure interoperability and reuse between diverse platforms and applications.

The MHEG standard provides both referencing and encapsulation of monomedia data through standard referencing mechanisms. Data encoding identification may be achieved via a standard MHEG catalogue, private catalogues or appropriate externally defined catalogues.

Composition of multiple media

In order to provide true multimedia presentation (as opposed to merely presentation of multiple media types), the coding scheme must support facilities to relate individual media objects to each other in order to produce meaningful output. This is addressed through the composition function with relationships between media objects being expressed in the parent object (or script).

Various techniques exist for the specification of relationships between objects within a multimedia composition. These techniques are often tightly tied to the nature of the editing and presentation environment or to the requirements of specific applications. In particular, popular techniques that are employed range from basic spatial composition using drag-and-drop interfaces and basic temporal composition using timeline editing, through to complex multimedia scripting languages. Scripting languages typically offer near-natural language or visual specification of complex application logic and presentation control with tightly integrated hardware and software presentation sub-systems.

It is possible to identify a number of key capabilities to support advanced multimedia composition:

- specification of spatial relationships formed by explicit links between objects,
- specification of temporal relationships formed by explicit links between objects,
- dynamic control of the spatial, temporal and audible parameters associated with objects in the system,
- run-time access to the state of, and parameters associated with, multimedia objects in the system,
- run-time capture of events associated with multimedia objects in the system,
- run-time capture of external events such as user interaction,
- support for logical combinations, conditions and tests of object state, object parameters, variables, internal and external events,
- support for flow control within scripts such as conditional execution, branching and loops.

The MHEG standard aims to provide a unified structure for the expression of relationships between multimedia objects. This structure, which is contained in a composite object, can be used to define dynamic spatial and temporal relationships between the components of the composite (and other objects) and to provide run-time access and operations on component state and component parameters. The MHEG Part 1 standard provides only minimal support for script-like flow control and complex logical and arithmetic operations. Such functionality is to be provided later in the MHEG scripting language.

The structure expressing inter-object relationships takes the form of a generalised hypermedia link and comprises a condition/action pair known as a link. Sets of links can be attached to objects (known as source objects). Each link expresses:

- a set of conditions in the form of logical and arithmetic tests on object state, object parameters and events within the system — conditions within the link are logically combined to form a single condition,
- a set of actions each targeted at specific multimedia objects within the system — the actions take the form of operations affecting object state, object parameters and causing new events to be generated.

Events (state changes, parameter changes, external events) within the multimedia system cause the evaluation of all currently active link conditions. A link condition which is satisfied causes the corresponding actions to take place on the targeted objects as illustrated in Fig 8.

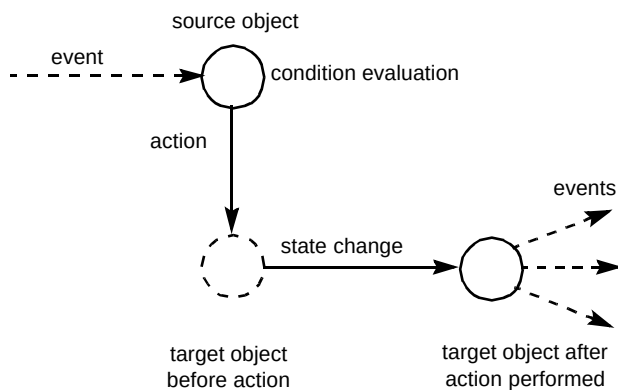


Fig 8 Link effect between objects.

Typical uses for the link condition/action structure are detailed in the following sections on spatial and temporal composition in which links may be used to achieve visual integration and synchronisation of media playback.

Spatial composition

Spatial composition (e.g. figure and caption, graphics overlay) is provided within an (X, Y) orthogonal co-ordinate system. Minimal facilities to support include:

- X, Y positioning (relative and absolute),
- scaling and clipping.

The MHEG standard provides an expressive set of mechanisms for defining the spatial relationships between media objects. These mechanisms may be used to produce complex visual effects and relationships between visual media objects such as scrolling, clipping, scaling, grouping of objects and windowing effects. Relationships are defined using link constructs between pairs of objects. A simplified view of the features supported by the standard is given in Fig 9 with links indicated by arrowed lines. The features illustrated comprise:

- X, Y positioning of a visible window on a media object relative to the parent object, sibling object or any other object within the presentation system,
- width, height specification of the visible window for clipping purposes,
- X, Y positioning of the media object relative to the visible window for clipping purposes,
- width, height specification of the media object for scaling purposes.

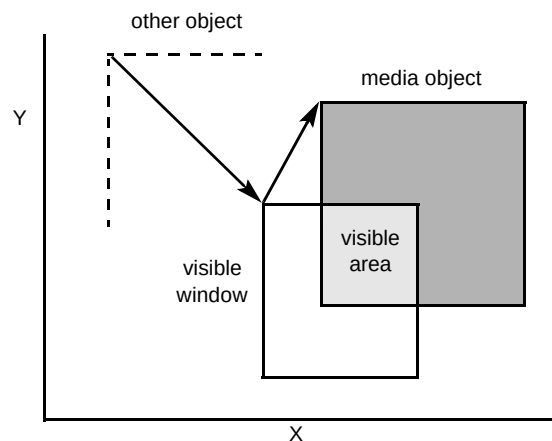


Fig 9 Simplified view of spatial composition.

Spatial composition may occur at many levels and can lead to complex visual presentations. Additionally, the standard considers Z-axis positioning and transparency in order to achieve overlay effects between media objects.

All attributes of spatial composition may be controlled at run time with actions attached to links allowing dynamic

visual effects to be realised. Typical action effects include altering the position and size of the visible window or media object.

Temporal composition (synchronisation)

Temporal composition (e.g. audio commentary to slide-show, video subtitling) is provided along a single dimension co-ordinate system representing time. Facilities to be supported include:

- start/stop times for media playback,
- positioning/repetition of playback within media,
- speed variation.

The MHEG standard again provides an expressive set of mechanisms for defining the temporal relationships between media objects. These mechanisms may be used to control the relative speed, playback position and duration of media objects in addition to defining complex synchronisation effects. Relationships are defined using link constructs between pairs of objects. A simplified set of examples of synchronisation facilities supported by the standard are illustrated in Figs 10 to 14 with links indicated by arrowed lines.

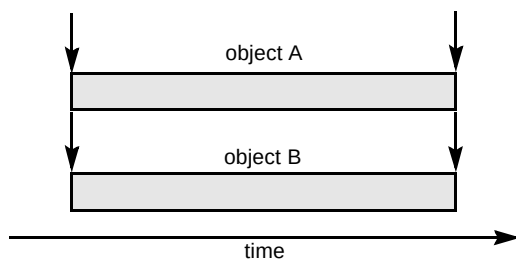


Fig 10 Parallel playback of objects.

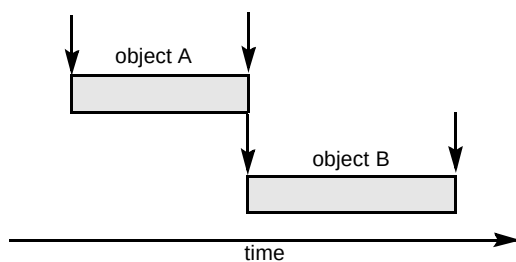


Fig 11 Sequential playback of objects.

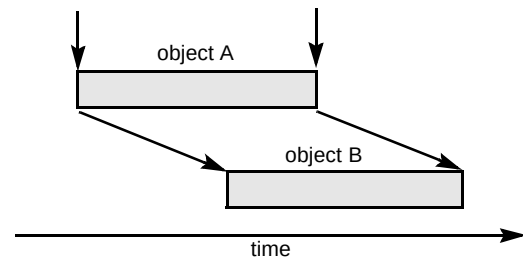


Fig 12 Overlapped playback of objects.

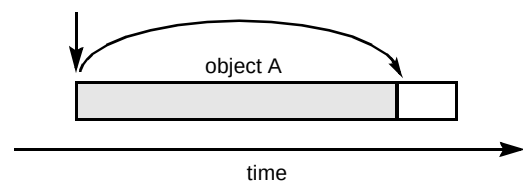


Fig 13 Playback for specified duration.

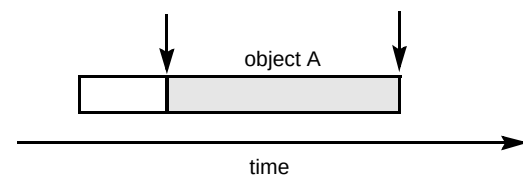


Fig 14 Playback from specified position.

Temporal composition may occur at many levels and can lead to complex synchronisation effects between many objects.

Temporal composition is provided for media objects with implicit duration (e.g. audio-visual sequences) and for media objects with no implicit duration (e.g. text, images) where the length of presentation is to be controlled.

All parameters of temporal composition may be controlled at run time within an MHEG object allowing complex synchronisation to be realised. A typical action effect is to vary the speed of playback, start times and stop times for media objects.

5.3 Referencing

The ability to accurately and uniquely reference objects within a multimedia system is a key requirement for the support of reuse and composition of multimedia data structures. The following facilities can be identified:

- referencing of media data on remote and local storage sub-systems — references must be sufficiently powerful and general to deal with a wide range of platforms, networks, filing systems and storage devices on which media data may be stored (note that this includes real time media devices, e.g. cameras, microphones, in addition to traditional storage devices),
- referencing of storage instances stored on remote and local storage sub-systems — the ability to uniquely identify storage instances is essential for reuse,
- referencing of presentable instances within the presentation environment by the using application and by other objects — each presentable instance within the presentation environment must have a unique identifier to allow the application or other objects to request preparation, presentation or removal of the object and associated resources,
- referencing of run-time instances of multimedia objects within the presentation environment by the using application and by other objects — each run-time instance must be uniquely identified to allow the using application or other objects to control the presentation of that object,
- referencing of the components of multimedia composition — each component within a composite object should be addressable relative to the composite object; components should be addressable in their storage, presentable and run-time states.

The requirement for referencing of multimedia objects and the components of a multimedia composition across storage, presentable and run-time states together with media data is illustrated in Fig 15.

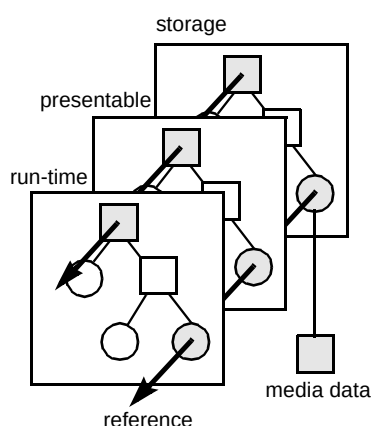


Fig 15 Referencing in a multimedia system.

Referencing of component objects relative to a parent (or ancestor) object is illustrated in Fig 16.

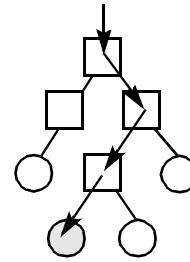


Fig 16 Referencing a component object.

The MHEG standard provides support for all the referencing techniques identified above and relies heavily upon them to achieve multimedia data structure reuse, composition and complex presentations.

5.4 Interaction

Multimedia applications are typically highly interactive and the requirement to reuse multimedia data content extends across interaction with the user as well as presentation to the user. A number of facilities can be identified which enable the concepts of interaction to be integrated within a multimedia coding scheme:

- interaction objects which may be mapped to typical modes of user input (text input, menus, buttons, pointers, check boxes, scroll bars, gauges),
- integration of interaction objects within the composition process in terms of spatial and temporal relationships with other objects,
- interrogation of the state of interaction objects — links representing logical relationships with other objects may be activated dependent upon the state of an interaction object,
- interrogation of the state of interaction objects by the using application in order to pass user input to the application for application specific processing.

The ability to support interaction objects within the composition structure allows the definition of reusable user interface components which may be applicable to a wide range of applications.

The MHEG standard supports the composition of interaction objects within multimedia data structures and enables the definition of reusable user interface components which are independent of application. Composite objects may contain links which perform interrogation and logical testing of the state of interaction objects and cause appropriate actions to occur.

5.5 Example multimedia object

This section illustrates the generic multimedia concepts outlined above through the construction of an example multimedia object. The example is intended to clarify the concepts discussed but is not intended to represent a particular multimedia coding scheme (e.g. MHEG).

The example multimedia object will represent a commonly used sub-component of a multimedia presentation — a subtitled audio-visual clip with VCR-like user control. This example provides an opportunity to illustrate:

- referencing of monomedia data,
- encapsulation of monomedia data,
- referencing of monomedia components,
- composition of a multimedia structure,
- spatial relationships,
- temporal relationships,
- interaction facilities,
- progression through the life cycle.

Construction and presentation of the example multimedia object is illustrated in the following five phases.

Phase 1 — Fig 17 illustrates the referencing and encapsulation of monomedia data within simple multimedia component objects. In this example, large files such as video data and image data are referenced from the component objects while small files such as text subtitles are encapsulated within the component objects.

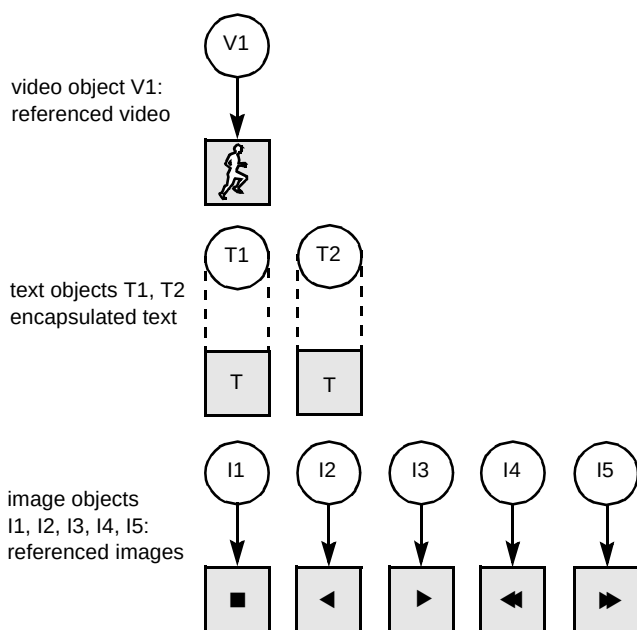


Fig 17 Referencing and encapsulation of media data.

Phase 2 — Fig 18 illustrates the composition of text and video components within a composite object representing the subtitled video and image components within a composite object representing the VCR controls. In this case, the monomedia components are referenced by the containing composite objects.

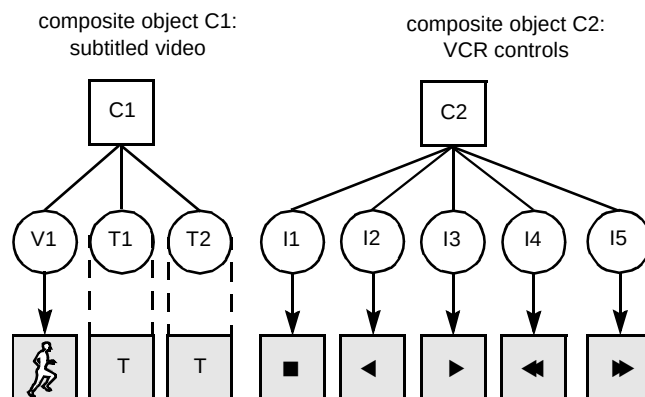


Fig 18 Composition of component objects.

Phase 3 — Fig 19 illustrates the composition of the subtitled video object and the VCR control object within an overall composite object for presentation purposes. It should be noted that as the subtitled video object is referenced by the higher level composite, it would be possible to exchange the subtitled video with another of similar structure. This allows the VCR control object to be used to control any number of subtitled video objects. The MHEG standard provides facilities to plug and unplug sub-structures from the composition to achieve such effects.

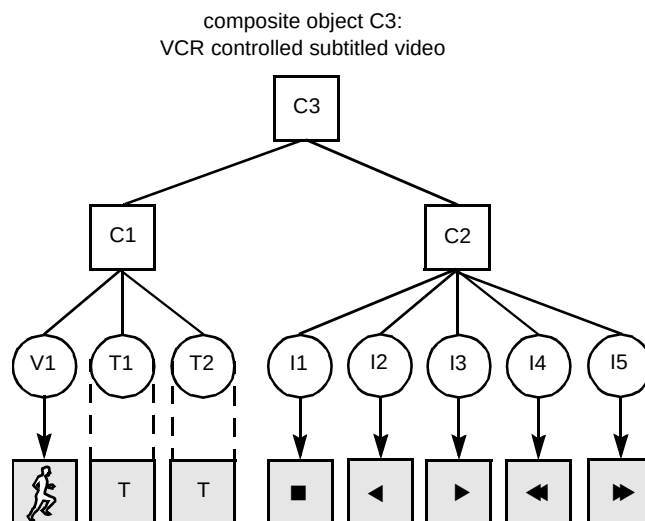


Fig 19 Higher levels of composition.

Phase 4 — Figs 20a to 20c illustrate the construction of a simplified control script to be attached to the composite objects within the structure. This structure takes the form of

a number of condition/action pairs which are used to monitor events in the system and take action as required.

Fig 20a shows actions to be taken when a using application requests the preparation of the composite object structure for presentation. The specified controls cause the preparation of sub-components within the hierarchy and the setting of initial position, size, temporal position and speed values for each of the monomedia components. In addition, the relative positions of the composites C1 and C2 with respect to the composite C3 are specified. The resulting spatial relationships are as shown.

Source	Condition	Target	Action
C3	Prepared	C1	Prepare
		C2	Prepare
		C1	Set Position 100, 100
		C2	Set Position 100, 420
C1	Prepared	C1.V1	Set Position 0, 0
		C1.V1	Set Size 400, 300
		C1.V1	Set Time 0
		C1.V1	Set Speed 0
		C1.T1	Set Position 200, 250
		C1.T1	Set Size 200, 50
		C1.T2	Set Position 200, 250
		C1.T2	Set Size 200, 50
C2	Prepared	C2.I1	Set Position 0, 0
		C2.I1	Set Size 80, 80
		C2.I2	Set Position 80, 0
		C2.I2	Set Size 80, 80
		C2.I3	Set Position 160, 0
		C2.I3	Set Size 80, 80
		C2.I4	Set Position 240, 0
		C2.I4	Set Size 80, 80
		C2.I5	Set Position 320, 0
		C2.I5	Set Size 80, 80

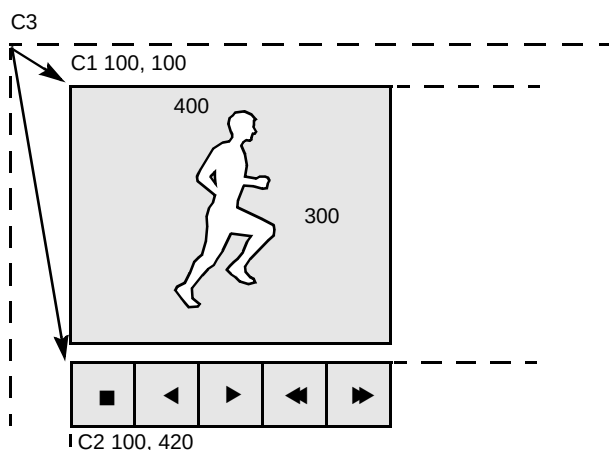


Fig 20a Specification of initial spatial relationships.

Fig 20b shows actions to be taken when a using application requests that the composite object structure be

presented. The specified controls cause the start of sub-components within the hierarchy, resulting in presentation of the video and VCR controls.

Source	Condition	Target	Action
C3	Started	C1	Start
		C2	Start
C1	Started	C1.V1	Start
C2	Started	C2.I1	Start
		C2.I2	Start
		C2.I3	Start
		C2.I4	Start
		C2.I5	Start

Fig 20b Presentation of composite object.

Fig 20c shows actions to be taken during the presentation of the composite object structure. In particular, actions are present to control the playback of the text subtitles at the correct points in the video with the timeline synchronisation as shown. Additionally, actions are provided to control the speed of video playback via the VCR control buttons which may be selected by the user.

Source	Condition	Target	Action
C1.V1	Time=5	C1.T1	Start
	Time=15	C1.T1	Stop
	Time=25	C1.T2	Start
	Time=30	C1.T2	Stop
	Stopped	C1.T1	Stop
	Stopped	C1.T2	Stop
C2.I1	Selected	C1.V1	Set Speed 0
		C1.V1	Stop
C2.I2	Selected	C1.V1	Set Speed -100
		C1.V1	Start
C2.I3	Selected	C1.V1	Set Speed 100
		C1.V1	Start
C2.I4	Selected	C1.V1	Set Speed -500
		C1.V1	Start
C2.I5	Selected	C1.V1	Set Speed 500
		C1.V1	Start

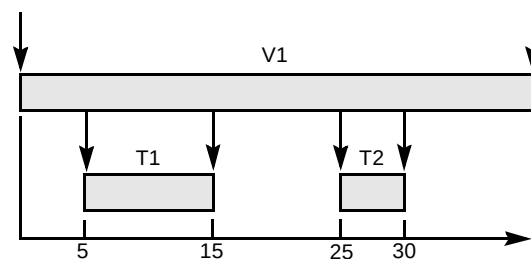


Fig 20c Specification of temporal synchronisation and control.

Phase 5 — Fig 21 illustrates the production of two separate run-time instances (e.g. at the request of an application) of the same composite object structure. It should be noted that as every spatial and temporal positioning specified within the composite object structures is relative, then only positioning of the outermost object C3 is required.

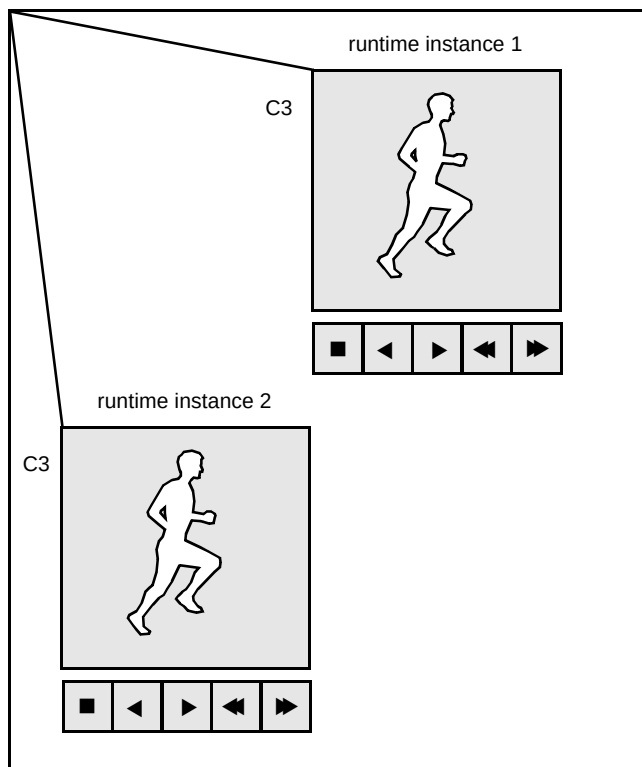


Fig 21 Creation of run-time instances.

6. Multimedia architecture

The advent of reusable multimedia content, which is independent of storage, distribution, presentation platform and application, implies a change in the nature of multimedia architectures. As previously indicated, existing architectures blur the distinction between content and application and rely on 'stove-pipe' hardware and software systems. These systems are highly integrated and dependencies exist between all layers from application to hardware. Newer architectures are beginning to rely on the development of standardised interfaces between key components of the architecture. Standardised multimedia information content allows this process to move a step further.

The following are some of the key components to be found in a multimedia architecture developed around reusable multimedia content. A simplified architecture

together with potential standardised interfaces between the components is illustrated in Fig 22.

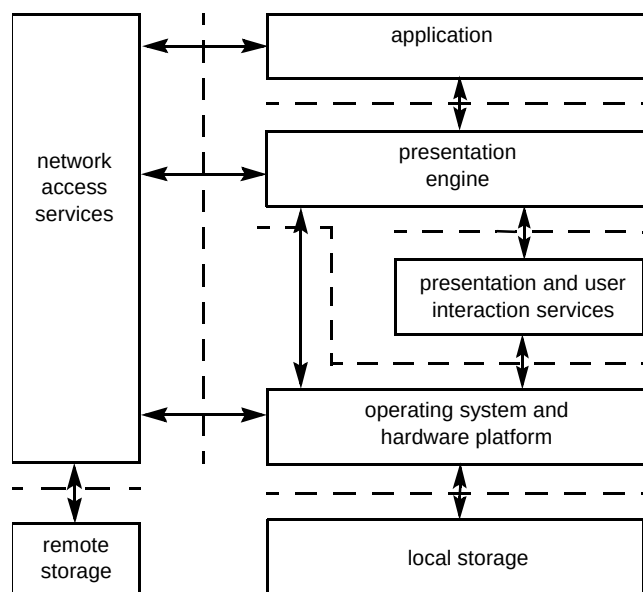


Fig 22 Example multimedia object with interaction.

The architecture comprises:

- the application, providing context specific processing and interfacing to the underlying presentation engine — the application is able to control the presentation and interrogation of multimedia objects through this interface,
- the presentation engine, providing processing of multimedia objects encoded according to the standard coding scheme — the presentation engine has access to remote and local storage services through the network and local hardware platforms respectively, and is responsible for the processing of the composition rules expressed within the multimedia objects and for issuing requests to the underlying presentation services,
- the presentation and user interaction services, providing media presentation and user interface facilities through the underlying operating systems and hardware platform,
- local and remote storage, providing storage for multimedia content and media data,
- network services, providing access to remote storage and advanced facilities such as object location services and quality of service negotiation.

A key requirement is the definition of standard interfaces between the application and presentation engine and between the presentation engine and presentation

services. Definition of these interfaces will allow the development of 'black-box' presentation engines that are able to process multimedia content encoded according to the standard coding scheme. Development of such engines for a wide variety of platforms will enable inter-application and inter-platform multimedia communications on a wide scale.

7. Conclusions

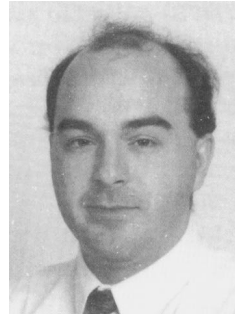
This paper has briefly presented some of the basic concepts associated with complex multimedia presentation and the requirements for a generic and standardised multimedia coding scheme. In addition, architectural considerations for future multimedia systems have been presented. Particular reference has been made to the emerging ISO MHEG (Multimedia and Hypermedia Information Coding Expert Group) standard as an example of how such concepts and architectures may be provided.

It is anticipated that standardised coding of multimedia content will become a key technology with the emergence of mass-market multimedia applications and services. In particular, the costs associated with multimedia content production are expected to highlight the need for a widespread coding scheme allowing content reuse.

It is clear that content providers, service providers, application developers, network providers, software and hardware manufacturers all need to consider these issues if widespread uptake of multimedia applications and services is to be achieved.

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First employed by BT Laboratories in 1989 he worked on the development of multimedia architectures based on object oriented database management systems. In 1989 he became a member of the ISO MHEG group developing a standardised multimedia encoding and has been involved in implementation of the standard at BTL.

He is currently a member of the business products solution design unit and is working on evolution planning for multimedia on-line services to the residential and business environments.